

Python and Table Readiness

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can specify a table schema.

I can check types and units as well as names.

I can distinguish structural validity from truth.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Python variables, types, expressions, rows, and columns give a reproducible representation of data and transformations.



LEARNING TARGET Trace mean for [4,8,12] using `sum(values)/len(values)`. What should the program do for []? Explain why returning 0 silently is misleading.

Builder name and experiment date:

Readiness for the connected explanation

What does `row["used"]` mean?

Readiness for File input and output

Is "12" immediately usable as an integer count?

My readiness explanation

Readiness for Exceptions and recovery

Which error can `int("twelve")` raise?

My readiness explanation

This week's end goal

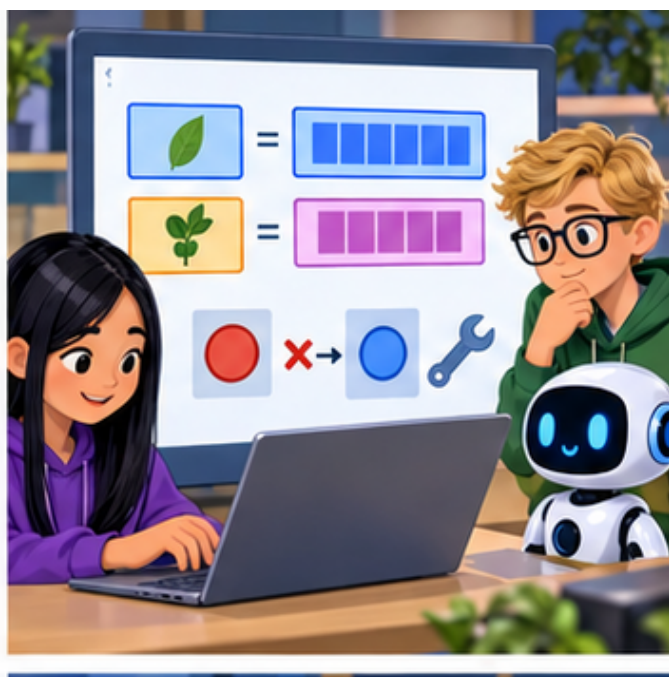
By the end, I can explain python and table readiness using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use python and table readiness to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A fictional classroom logs the duration of each activity. Before comparing durations, define a schema:

activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session. These decisions make data interpretable.

Apply the connected idea: File input and output

Use a teacher-provided local sandbox folder, never a real personal file. Create durations.txt with two lines: 12 and 8. The goal is to read the text, convert it to integers and report the total. Keep the original file unchanged for this activity.

Apply the connected idea: Exceptions and recovery

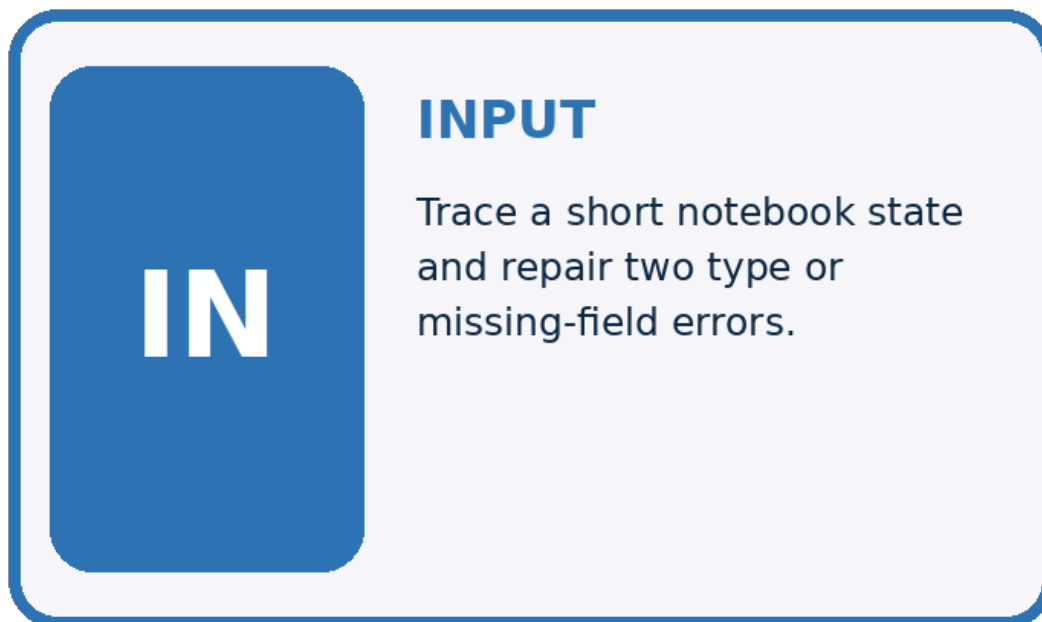
A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60. Invalid numeric text should return a clear status rather than crash or pretend that the duration was zero. Test both the successful and failing paths.

3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

Trace the information, fitted behavior, and evaluation proof separately.





ML

MODEL

Python variables, types, expressions, rows, and columns give a reproducible representation of data and transformations.



OUT

EVIDENCE

Trace a short notebook state and repair two type or missing-field errors.

QUESTION Trace a short notebook state and repair two type or missing-field errors.

MODEL IDEA Python variables, types, expressions, rows, and columns give a reproducible representation of data and transformations.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD `widths = [4.2, 5.0, 6.1]` stores three decimal measurements. `mean_width = sum(widths) / len(widths)` evaluates to 5.1.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

```
rows = [{"activity": "Reading", "minutes": 12},
        {"activity": "Sorting", "minutes": 0}]

for row in rows:
    valid = (type(row["activity"]) is str
             and len(row["activity"]) > 0
             and type(row["minutes"]) is int
             and 0 <= row["minutes"] <= 60)

    print(valid)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: activity

Evidence: Nonempty text

Meaning: Names the session activity

Step or case: minutes

Evidence: Integer 0 through 60

Meaning: Duration in minutes

Read each row from left to right and connect the evidence to the stated rule.

The worked script prints True twice. `type(value)` is int checks the exact type here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries. The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written schema.

Check a row in separate steps

A schema is a checklist for what a row must contain. First ask whether the activity name is text and is not empty. Next ask whether minutes is an integer. Then ask whether that integer lies between 0 and 60, including both ends. Keep these checks separate when explaining a rejection. The number 12 and the text "12" look similar on a page, but only the number meets this lesson's minutes rule.

Apply the connected idea: File input and output

```
with open("durations.txt", "r", encoding="utf-8") as handle:
    lines = handle.read().splitlines()
    values = []
    for line in lines:
        values.append(int(line))
    print(sum(values))
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Local file

Evidence: 12 then 8 on separate lines

Meaning: Read as text

Step or case: Conversion

Evidence: "12" to 12 and "8" to 8

Meaning: Integer values

Step or case: Summary

Evidence: 12 + 8

Meaning: Print 20

Read each row from left to right and connect the evidence to the stated rule.

The mode `r` means read. `with` closes the file when the block finishes, including when an error occurs. `splitlines` separates lines without retaining newline characters. This example assumes each line is a valid integer text; an empty line or `twelve` raises `ValueError` during `int` conversion.

Separate file text from numbers

The file contains characters, not ready-made Python integers. Reading gives text; `splitlines` separates the lines; `int` converts each valid line into a number. The lines `"12"` and `"8"` become 12 and 8, so their total is 20. Keep those three stages visible in your trace. A line such as `"twelve"` fails conversion, and a missing file is a different problem. Neither failure means the observed duration was zero.

Apply the connected idea: Exceptions and recovery

```
def parse_minutes(text):
    try:
        value = int(text)

    except ValueError:
        return {"status": "invalid text", "value": None}

    if value < 0 or value > 60:
        return {"status": "out of range", "value": None}

    return {"status": "ok", "value": value}

for text in ["12", "zero", "0", "61"]:
    print(parse_minutes(text))
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: "12" or "0"

Evidence: Conversion and range pass

Meaning: ok with numeric value

Step or case: "zero"

Evidence: Conversion fails

Meaning: invalid text with None

Step or case: "61"

Evidence: Integer outside range

Meaning: out of range with None

Read each row from left to right and connect the evidence to the stated rule.

try covers the operation expected to fail; except ValueError handles that known conversion failure. Returning early ends the failing path. This contract expects a string: it does not claim to handle every possible object. Spaces around an integer are accepted by int, and that behavior should be documented.

Give each failure an honest result

Trace the function as a decision path. First try to turn the text into an integer. If that conversion raises `ValueError`, return invalid text and `None`. If conversion succeeds, check the range. A value outside 0 through 60 returns out of range and `None`. Only a value passing both stages returns ok and its number. `None` means there is no accepted number; it must not be added to a duration total.

Connect the picture and the explanation

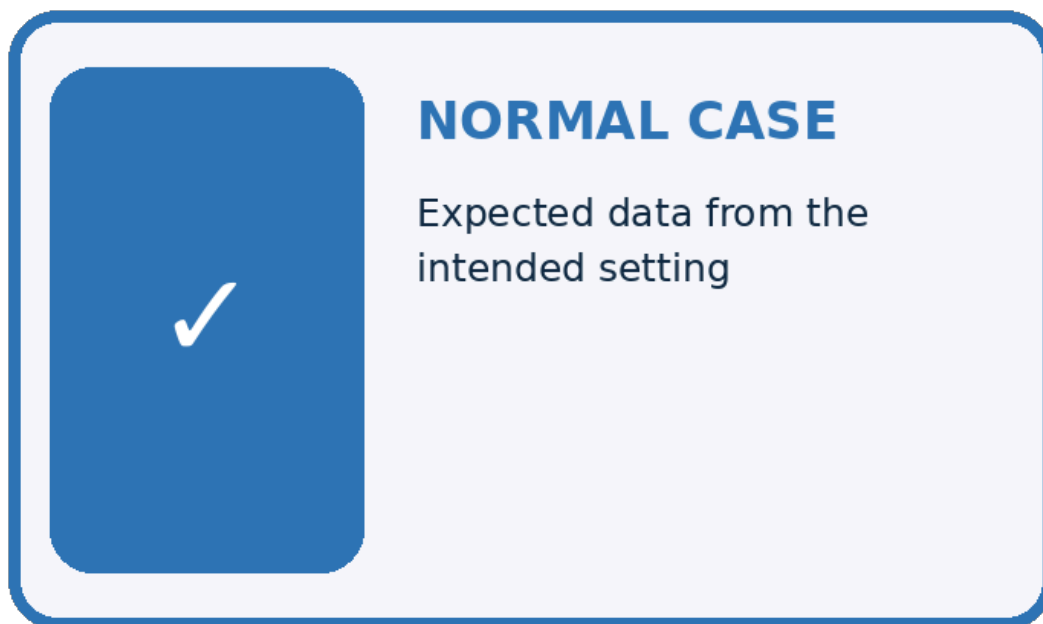
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





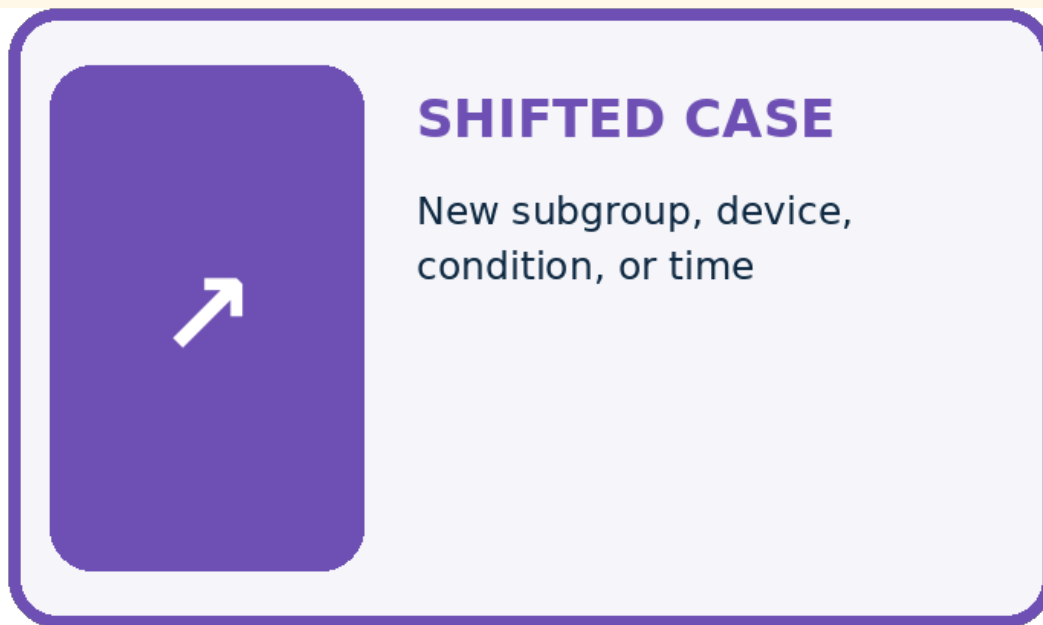
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: S changes from 0 to None.

Prediction: Mean becomes $(2+6)/2=4$ cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

VARIABLE

name bound to a value

2

DATA TYPE

kind of value and valid
operations

3

DATAFRAME

table represented in code

4

EXPRESSION

code that computes a value

Term · Precise meaning · My ML example

Term: variable

Precise meaning: name bound to a value

My ML example: _____

Term: data type

Precise meaning: kind of value and valid operations

My ML example: _____

Term: dataframe

Precise meaning: table represented in code

My ML example: _____

Term: expression

Precise meaning: code that computes a value

My ML example: _____

MISCONCEPTION TO REPAIR If notebook cells eventually run, execution order and data types do not matter.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Schema

Meaning: Rules for the fields and values in records

Word: Type

Meaning: A category such as integer or text

Word: Unit

Meaning: The quantity scale attached to a number

Word: Constraint

Meaning: A rule a valid record must satisfy

Terms for File input and output

Term · Meaning

Term: File

Meaning: Stored data with a name and location

Term: Path

Meaning: A location used to find a file

Term: Encoding

Meaning: A rule for representing text as bytes

Term: Context manager

Meaning: A with block that manages a resource such as closing a file

Terms for Exceptions and recovery

Term · Meaning

Term: Exception

Meaning: A signal that an operation could not complete normally

Term: Handler

Meaning: Code that responds to a specified exception

Term: Recovery

Meaning: A defined next step after a failure

Term: Status

Meaning: A label stating the outcome of a processing step

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 4 • WORKED EVIDENCE

- 1 widths = [4.2, 5.0, 6.1] stores three decimal measurements.
- 2 mean_width = sum(widths) / len(widths) evaluates to 5.1.
- 3 Adding 'unknown' to the numeric list causes a type problem.
- 4 Clean or explicitly encode values, rerun top to bottom, and record the result.

STEP 1 widths = [4.2, 5.0, 6.1] stores three decimal measurements.

STEP 2 mean_width = sum(widths) / len(widths) evaluates to 5.1.

STEP 3 Adding 'unknown' to the numeric list causes a type problem.

STEP 4 Clean or explicitly encode values, rerun top to bottom, and record the result.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Would minutes of 61, "12" and 12 each pass? Explain the type distinction.

Apply the connected idea: File input and output

If the file contains 0, 3 and 7 on separate lines, predict the values and total.

Apply the connected idea: Exceptions and recovery

Predict statuses and values for "-1", " 5 " and "3.5".

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Python and Table Readiness

Inspect a small table before asking Python to compute an average.

The facts we will use

Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

AI ACADEMY / AI-4/4

Python and Table Readiness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check records

Four rows contain three measured values and one missing value.



2. Calculate mean

Measured sum is $2+6+0=8$; mean is $8/3$ cm, approximately 2.67 cm.



3. Report coverage

Report 3/4 measured, retaining the missing record separately.

CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Check records

What happens and why: Four rows contain three measured values and one missing value.

Step: 2. Calculate mean

What happens and why: Measured sum is $2+6+0=8$; mean is $8/3$ cm, approximately 2.67 cm.

Step: 3. Report coverage

What happens and why: Report 3/4 measured, retaining the missing record separately.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

The measured-row mean may not represent the missing plants; investigate why those readings are absent.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: If notebook cells eventually run, execution order and data types do not matter. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE If notebook cells eventually run, execution order and data types do not matter.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A row can satisfy every schema rule and still be a wrong observation. Schema validation checks our stated structure and constraints, not whether someone measured honestly or selected a representative sample.

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

Apply the connected idea: File input and output

File reading does not establish that the values are valid durations. After conversion, apply the schema and record any rejected lines. Do not upload personal records to an external tool for this exercise.

The file contains twelve on one line. Why should the program not silently turn it into zero?

Apply the connected idea: Exceptions and recovery

Recovery must fit the task. A caller may ask for corrected input or exclude the row with a recorded reason. It must not use None as a number or claim that rejected records were successfully processed.

A batch has 8 valid and 2 invalid lines. What should its report include?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a column alone does not convert values.

Apply the connected idea: File input and output

If Python cannot find durations.txt, it raises FileNotFoundError. Check the current working folder and exact filename. Do not switch to write mode to solve a missing input: opening an existing file with w would replace its contents.

Apply the connected idea: Exceptions and recovery

A bare except that returns 0 hides unrelated programming errors and corrupts missing values. Catch the expected exception, preserve an error status and keep the unknown value as None. If file reading is added, handle FileNotFoundError separately with an actionable message.

Practise with less help: Classify records

Use workbook W2 for the connected example “Rows columns and schemas”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: File input and output

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Practise the connected topic: Exceptions and recovery

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Notebook Detective: predict five variable values, trace two table filters, repair two errors, and rerun from a clean state.

- 1. Write the hypothesis and expected evidence before running.
- 2. Notebook Detective: predict five variable values, trace two table filters, repair two errors, and rerun from a clean state.
- 3. Run one normal, one boundary, one missing, and one shifted case.
- 4. Calculate the metric from raw counts or row-level errors.
- 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

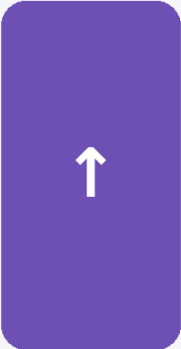
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: variable, data type, dataframe, expression.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE If notebook cells eventually run, execution order and data types do not matter.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

THIS WEEK'S CHECKPOINT Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Trace a short notebook state and repair two type or missing-field errors.

REQUIRED PROOF Trace a short notebook state and repair two type or missing-field errors.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Inspect Before Modeling.

13. Study a complete case

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Python `lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).`

The question

Trace and handle an empty list.

Worked explanation

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without observations.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace the calculation and its empty case

Evidence or stage · Value or result

Evidence or stage: lengths

Value or result: [4,6,8]

Evidence or stage: sum

Value or result: 18

Evidence or stage: count

Value or result: 3

Evidence or stage: mean

Value or result: 6.0

Evidence or stage: empty list

Value or result: No mean, guard division by zero

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Trace and handle an empty list.

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without observations.

14. Try a new case independently

Independent case: Changed independent case

Supplied values are 3,7,14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Trace Python: `lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3`. Give selected and total.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Trace mean for [4,8,12] using `sum(values)/len(values)`. What should the program do for []? Explain why returning 0 silently is misleading.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.